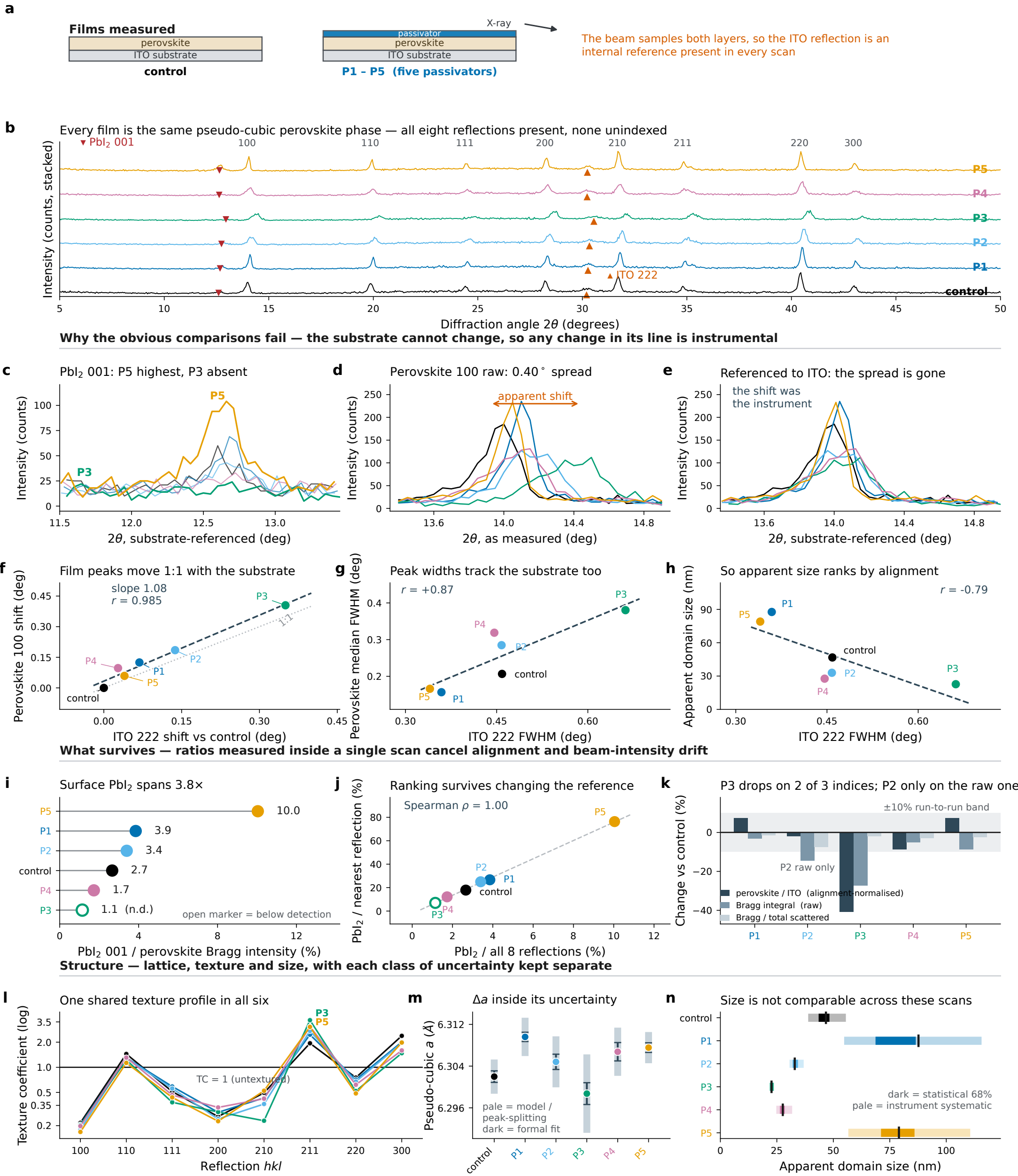


Five surface passivators on perovskite films, screened by X-ray diffraction

One scan per film (n = 1, no repeats). Identical protocol throughout: Cu K α λ = 1.54184 Å, 5–50 ° 2 θ , 0.05 ° step, 0.3 s per step, Bragg–Brentano geometry, monochromator on.



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Phase identity — same pseudo-cubic cell
PbI₂ / perovskite Bragg ratio
Comparative crystallinity indices
Texture coefficients
Lattice parameter Δa
Crystallite size ΔD
Absolute degree of crystallinity
Microstrain

VALID	all 8 reflections index in all six films
VALID	within-scan ratio; ranking robust (ρ = 1.00)
VALID	protocol identical across all six scans
VALID	fixed convention; substrate reflection excluded
NOT COMPARABLE	peak shifts track the substrate 1:1 (r = 0.985)
NOT COMPARABLE	peak widths track the substrate (r = +0.87)
BOUNDED RANGE ONLY	amorphous halo degenerate with background
SUPPRESSED	Williamson–Hall slope negative in 5 of 6